

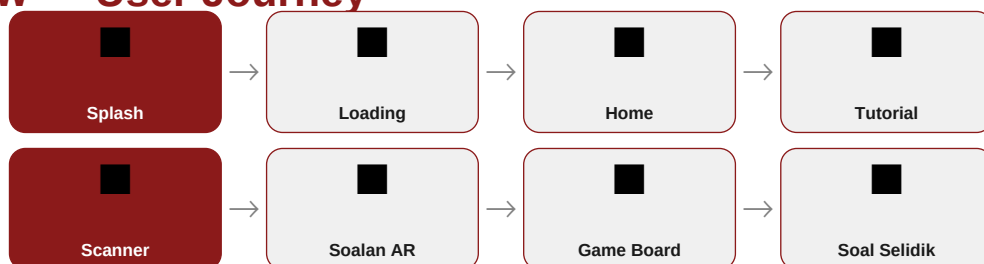


Interactive Game Board with AR — Melaka Theme

App Screens Overview & Progress Report Phase 1 & Phase 2 — June 2026

Platform	Android (Flutter 3.x, Dart 3.x)
AR Engine	ARCore via ar_flutter_plugin
Min SDK	Android API 24 (ARCore required)
Backend (planned)	Flask API + SQLite
Total Screens	9 screens implemented
Build	GitHub Actions → Release APK

App Flow — User Journey



9 Screens Built	GitHub Actions	AR Markers	Soal Selidik
Fully implemented all routes	Auto-build APK on every push	11 Melaka landmarks	3-section survey with validation

All Screens — Detailed

1. Splash Screen

Splash Screen

Route: / (entry point)



- Logo i.-GB muncul dengan animasi bounce (1.10x scale)
- Star burst effect bila user tap logo (bintang terbang keluar)
- Sound effect — blinkingstar.mp3 dimainkan
- Auto-navigate ke Home screen selepas animasi selesai

Tech:

- AnimationController (bounce), AudioPlayer, Navigator.pushNamed("/home")

2. Loading Screen

Loading Screen

Route: /loading



- Animasi coin-flip pada logo (rotateY 360°)
- Bintang kecil terbang keluar dari tepi logo
- Bintang bermula dari tepi logo, saiz kecil, terbang ke luar sahaja
- Logo topic di-flip semasa loading
- 5 topic image cards diload (Sejarah, Budaya, Pelancongan, dll)

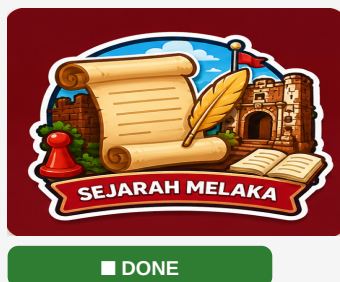
Tech:

- TweenAnimationBuilder, Transform.rotate, Stack overlay

3. Home Screen

Home Screen / Main Menu

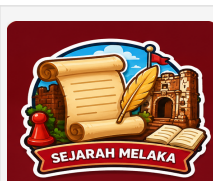
Route: /home



- Input nama pemain (TextField + validation)
- Alert dialog jika nama kosong bila mula
- 5 Topic Cards dengan imej:
 - ■ Sejarah Melaka (topic_sejarah_melaka.png)
 - ■ Budaya (topic_budaya.png)
 - ■ Pelancongan (topic_pelancongan.png)
 - ■ Matematik (topic_matematik.png)
 - ■ Seni Bina (topic_seni_bina.png)
- Buttons: Tutorial | Nota | About
- Score badge ditunjuk (nama pemain)

Navigates to:

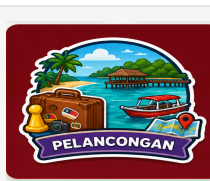
- Scanner screen dengan args: {playerName, topic}



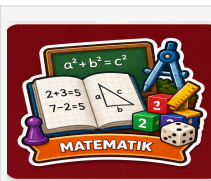
Sejarah



Budaya



Pelancongan



Matematik



Seni Bina

↑ 5 Topic Image Cards yang ada dalam Home Screen

4. Tutorial Screen

Tutorial ScreenRoute: /tutorial



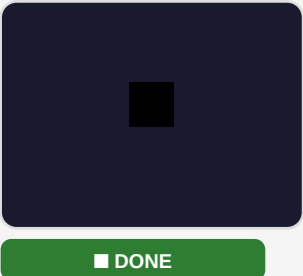
- 4-step tutorial dengan PageView (swipe kiri-kanan)
- Step 1: "Imbas Papan Permainan" — animasi scan illustration
- Step 2: "Kad Tempat Muncul" — animasi flip card
- Step 3: "Jawab Soalan" — MCQ illustration
- Step 4: "Kumpul Ganjaran" — trophy illustration
- Progress dots di bawah
- Button "Mula" di step terakhir → ke Scanner

Illustrations:

- _PhoneScanIllustration, _FlipCardIllustration (custom drawn)

5. Scanner / AR Camera Screen

Scanner Screen (AR)Route: /scanner



- ARCore camera view (live camera feed)
- Viewfinder overlay — 4 bracket corners (custom painter)
- 11 AR image markers didaftarkan:
 - A'Famosa, Cheng Hoon Teng, Christ Church,
 - Jonker Street, Masjid Selat, Menara Taming Sari,
 - Muzium Kapal Selam, Stadthuys, St. Paul's Hill,
 - Beca Melaka, Tempat Melaka
- Soal Selidik button (top-right, pill putih)
- Back button (top-left, bulat hitam)
- Arahan scan di bahagian bawah
- Emulator fallback screen (kamera tidak tersedia)

On marker detected:

- → Trigger 3D Flip Card animation

6. AR Flip Place Card (dalam Scanner)



↑ Contoh 3 dari 11 gambar marker AR yang digunakan dalam Scanner

3D Flip Place Card

(popup dalam Scanner)



■ DONE

- Muka depan: Gambar tempat + nama landmark
- Auto-flip ke belakang selepas 1.8 saat
- Muka belakang: Quiz icon + soalan prompt
- Button "Teruskan ke Soalan" (amber)
- Background blur (ImageFilter.blur sigma=18)
- 3D flip animation (rotateY, perspective matrix)

Tap manual:

- User boleh tap kad untuk flip sendiri

7. Soalan AR Overlay (dalam Scanner)

Question Overlay (MCQ)

(overlay dalam Scanner)



■ DONE

- Header merah: emoji + nama landmark + topik badge
- Soalan ditunjuk (teks penuh)
- 4 pilihan jawapan (A, B, C, D) — animated buttons
- Colour feedback: hijau=betul, merah=salah
- Icon ✓ / ✗ selepas jawab
- Mesej "■ Betul! Markah ditambah!" atau "■ Salah"
- Button "Imbas Seterusnya" selepas jawab
- Scale animation (elasticOut) masa muncul

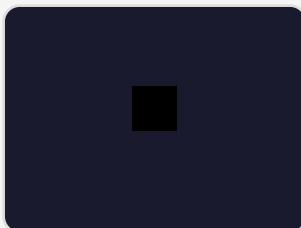
Questions data:

- lib/data/questions_data.dart — soalan untuk setiap topik

8. Soal Selidik Sheet (BARU)

Soal Selidik / Survey

(modal bottom sheet dari Scanner)



■ BARU DIBUAT

- Button "Soal Selidik" (pill putih, top-right scanner)
- Drag-up modal bottom sheet (DraggableScrollableSheet)

Bahagian A — Maklumat Responden:

- Q1: Status (Pelajar/Guru/Pelancong/Lain-lain)
- Q2: Peringkat Umur (7-12 / 13-17 / 18-25 / 26+)

Bahagian B — Pengalaman App:

- Q3: Kemudahan guna app (Sangat Mudah → Sukar)
- Q4: Pengalaman AR (Sangat Menarik → Tidak Menarik)
- Q5: Kesesuaian soalan (Sangat Sesuai → Tidak Sesuai)

Bahagian C — Penilaian:

- Q6: Star rating 1–5 bintang (interactive tap)
- Q7: Komen terbuka (TextField, optional)

■ Validation: semua * wajib sebelum boleh hantar

■ Thank you screen selepas submit

■■ NOTE: Soalan dummy — akak akan bagi soalan sebenar

9. Game Board Screen

Game Board (Snake & Ladder)

Route: /game



■ DONE

- Papan snake & ladder (zigzag klasik)
- Player token bergerak ikut skor
- Nama pemain dalam AppBar: "i.-GB — [Nama]"
- Square tiles dengan nombor + emoji landmark
- "Permainan Tamat!" popup dengan skor akhir
- Button: Kembali ke Menu | Main Semula

Scoring:

- Skor dari Scanner → move token di board

10. Skrin Sokongan

Nota Screen /nota	About Screen /about	AR Demo Screen /ar-demo
■ Rujukan / nota untuk pelajar	■ Info projek + namiahli kumpulan	■ Demo AR scene tanpa game flow

GitHub Actions — Auto Build APK

GitHub Actions Workflow

.github/workflows/build-apk.yml



■ DONE

- Trigger: push to main / manual "Run workflow"
- Flutter latest stable (auto — fixes Dart version)
- Java 17 (temurin)
- flutter pub get → flutter build apk --release
- Upload artifact: app-release-apk (30 hari)

How to download APK:

- GitHub → Actions tab → Build Release APK
- → Click run → Artifacts → Download app-release-apk

Fix applied:

- Removed flutter-version pin (was 3.32.1/Dart 3.8.1)
- Now uses latest stable → satisfies sdk: ^3.11.5

AR Scan Images — 11 Melaka Landmarks



A'Famosa



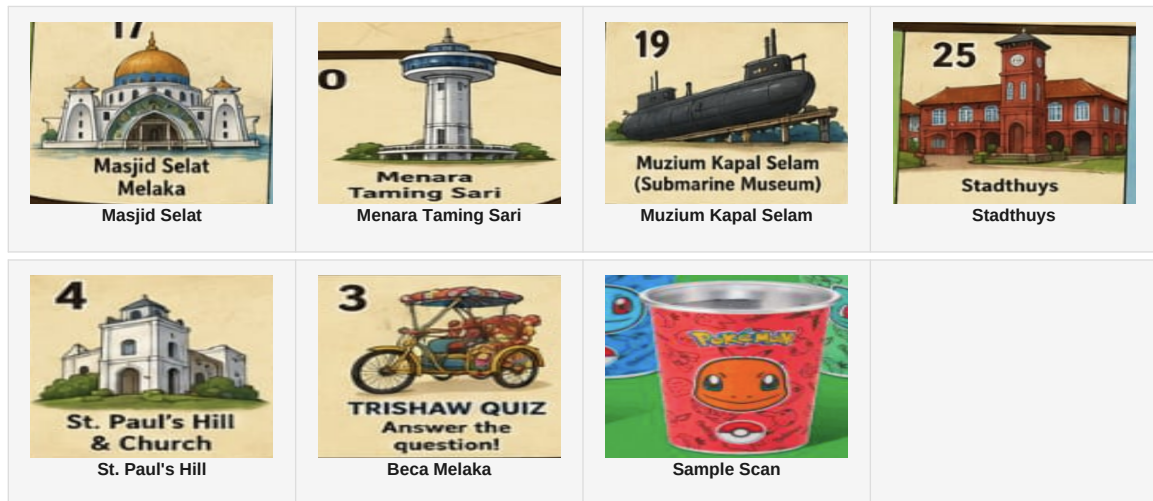
Cheng Hoon Teng



Christ Church



Jonker Street



Status Summary & What's Next

#	Feature / Screen	Status	Notes
1	Splash Screen	■ Done	Bounce + star burst + sound
2	Loading Screen	■ Done	Coin flip + flying stars
3	Home Screen	■ Done	5 topic cards + name input
4	Tutorial Screen	■ Done	4-step swipe tutorial
5	Scanner / AR Camera	■ Done	ARCore + 11 markers
6	Flip Place Card	■ Done	3D flip animation
7	MCQ Question Overlay	■ Done	Colour feedback + scoring
8	Soal Selidik Survey	■ Done	Dummy — akak akan update
9	Game Board (Snake & Ladder)	■ Done	Token movement + end screen
10	Nota Screen	■ Done	Reference notes
11	About Screen	■ Done	Team info
12	AR Demo Screen	■ Done	Demo tanpa game flow
13	GitHub Actions (APK build)	■ Done	Latest stable Flutter
14	Backend Flask API	■ Planned	Phase 3 — online leaderboard
15	SQLite score history	■ Planned	Phase 3
16	iOS support	■ Planned	ARKit needed
17	Real Soal Selidik questions	■ Pending	Akak bagi soalan → update